

The Bhattacharya–Rhoades sign-obstruction theorem: $q_e^{(n)}$ is not a $\mathbb{Q}$ -combination of fermionic slices of $SR_{n,r}$

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Abstract

Let $e \geq 2$, $n = ek'$ with $k' \geq 2$, and let $q_e^{(n)} := p_e^{k'} \in \Lambda_n$ be the composite- d virtual character. Let $SR_{n,r}$ denote the Bhattacharya–Rhoades wreath superspace [1] for $r \geq 1$, and let $M_{n,r}^{(k)}$ be its fermionic-degree- $(n-k)$ slice, viewed as an S_n -representation by restriction. We prove that

$$q_e^{(n)} \notin \text{span}_{\mathbb{Q}}\{\text{Frob}(M_{n,r}^{(k)}) : r \geq 1, 0 \leq k \leq n\}.$$

The obstruction is structural: the entire family of slice-Frobenii lies inside a subspace $W_n \subset \Lambda_n$ of dimension exactly n , which admits a clean description in the power-sum basis. Because $p_e^{k'}$ is supported at a single power-sum partition of length k' and $k' \geq 2$ forces multiple partitions of length k' to co-occur inside W_n , an explicit linear functional separates $q_e^{(n)}$ from the entire span.

The subspace W_n has codimension $p(n) - n$ in Λ_n ; for $n = 6$ this is $11 - 6 = 5$, matching the empirical null space found in the $(n, r) = (6, 3)$ probe. The empirical Schur-basis functionals L_1, L_2, L_3 from that probe are shown directly to lie in the annihilator of W_n .

1 Setup and statement

Fix $n \geq 1$ and $r \geq 1$. Write Λ_n for the space of homogeneous symmetric functions of degree n (over $\mathbb{Q}$), and use the standard bases $\{p_\mu\}$, $\{h_\mu\}$, $\{e_\mu\}$, $\{s_\lambda\}$ indexed by partitions of n . Let $\varepsilon(\mu) := (-1)^{n-\ell(\mu)}$ be the sign of the conjugacy class μ and $z_\mu := \prod_i i^{m_i(\mu)} m_i(\mu)!$ the centralizer order.

Let $\widetilde{\mathcal{OP}}_{n,r}^{(k)}$ denote the set of *r-ified ordered set partitions with k nonzero blocks*: an ordered decomposition $[n] = Z \sqcup B_1 \sqcup \cdots \sqcup B_k$ into a (possibly empty) zero block Z and k nonempty blocks $B_1, \dots, B_k$, together with a colouring of each element of B_i by $\{0, \dots, r-1\}$ and of each element of Z by $\{0, \dots, r-2\}$. This follows Bhattacharya–Rhoades [1, Sec. 3–4]; by their Theorems 4.23–4.24 the fermionic-degree- $(n-k)$ slice of the wreath-superspace module is

$$M_{n,r}^{(k)} \cong \mathbb{C}[\widetilde{\mathcal{OP}}_{n,r}^{(k)}] \otimes \det \quad \text{as } \mathbb{C}[\mathbb{Z}_r \wr S_n]\text{-modules.} \quad (1)$$

Restricting to $S_n \hookrightarrow \mathbb{Z}_r \wr S_n$ and taking Frobenius characteristic yields an element $\text{Frob}(M_{n,r}^{(k)}) \in \Lambda_n$.

Let

$$q_e^{(n)} := p_e^{k'} = p_{(ek')} \in \Lambda_n, \quad n = ek'. \quad (2)$$

This is the “composite- d ” virtual character studied in [2]; its Schur expansion for $(n, e) = (6, 3)$ is $p_3^2 = s_{(6)} - s_{(5,1)} + s_{(4,1^2)} + 2s_{(3,3)} - 2s_{(3,2,1)} + s_{(3,1^3)} + 2s_{(2^3)} - s_{(2,1^4)} + s_{(1^6)}$.

Theorem 1 (BR sign-obstruction). *Let $e \geq 2$ and $n = ek'$ with $k' \geq 2$. Then*

$$q_e^{(n)} \notin W_n^{\text{BR}} := \text{span}_{\mathbb{Q}} \left\{ \text{Frob}(M_{n,r}^{(k)}) : r \geq 1, 0 \leq k \leq n \right\}.$$

There is a further inclusion

$$W_n^{\text{BR}} \subseteq W_n := \bigoplus_{\ell=1}^n \mathbb{Q} \cdot P_\ell, \quad P_\ell := \sum_{\substack{\mu \vdash n \\ \ell(\mu)=\ell}} \frac{\varepsilon(\mu)}{z_\mu} p_\mu,$$

which is in fact an equality. In particular $\dim W_n^{\text{BR}} = n$, so the annihilator $(W_n^{\text{BR}})^\perp \subset \Lambda_n^$ has dimension $p(n) - n$. For any $\nu \vdash n$ with $\ell(\nu) = k'$ and $\nu \neq (e^{k'})$ the functional*

$$L_\nu(\phi) := \frac{z_{(e^{k'})}}{\varepsilon((e^{k'}))} [p_{(e^{k'})}] \phi - \frac{z_\nu}{\varepsilon(\nu)} [p_\nu] \phi \quad (3)$$

(where $[p_\mu]\phi$ extracts the p_μ -coefficient in the power-sum expansion) annihilates W_n^{BR} and satisfies $L_\nu(q_e^{(n)}) = z_{(e^{k'})} \varepsilon((e^{k'}))^{-1} \neq 0$.

Since $k' \geq 2$ and $e \geq 2$, the partition $\nu = (e+1, e^{k'-2}, e-1)$ is a legal witness (it is a partition of $n = ek'$ of length k' distinct from $(e^{k'})$).

The proof of Theorem 1 proceeds in three steps:

- (i) A generating-function identity for $\text{Frob}(M_{n,r}^{(k)})$ (Proposition 2).
- (ii) A power-sum evaluation of the resulting generating function via the classical plethystic identity $\log \bar{E} = \sum_k (-1)^{k-1} p_k/k$ (Proposition 3).
- (iii) A dimension count and length-support analysis of W_n (Proposition 4) that isolates the obstruction to $p_{(e^{k'})} \in W_n$.

2 The generating function for slice-Frobenii

Let $\bar{H} := \sum_{i \geq 0} h_i \in \Lambda[[x^{-1}]]$ (formally the specialisation of $H(t) = \sum h_i t^i$ at $t = 1$, understood as a graded infinite series whose homogeneous-degree- n piece $[x^n] \bar{H}$ is h_n). Similarly let $\bar{E} := \sum_{i \geq 0} e_i$. Note $\bar{H} \bar{H}' = (\bar{H} + \bar{H}') \cdot \text{something}$ is *not* what we use — we use $\bar{H}$ (resp. $\bar{E}$) purely as a generating-function device: expressions $(\bar{H}^r - 1)^k \bar{H}^{r-1}$ live in the free module $\prod_{n \geq 0} \Lambda_n$ and we take homogeneous parts to land back in Λ_n .

Proposition 2. *For all $n \geq 0$, $r \geq 1$, and $0 \leq k \leq n$,*

$$\text{Frob}(\mathbb{C}[\widetilde{\mathcal{OP}}_{n,r}^{(k)}]) = [x^n] (\bar{H}^r - 1)^k \bar{H}^{r-1} \in \Lambda_n \quad (4)$$

(without det-twist). Consequently

$$\text{Frob}(M_{n,r}^{(k)}) = [x^n] (\bar{E}^r - 1)^k \bar{E}^{r-1} = \sum_{j=0}^k \binom{k}{j} (-1)^{k-j} F_{r(j+1)-1}, \quad (5)$$

where $F_m := [x^n] \bar{E}^m$.

Proof. A configuration in $\widetilde{\mathcal{OP}}_{n,r}^{(k)}$ is determined by assigning each of the n elements to one of the $kr + (r-1)$ “position types” (B_i, c) for $1 \leq i \leq k$, $0 \leq c \leq r-1$, or (Z, c) for $0 \leq c \leq r-2$, subject to the constraint that each block B_i receives at least one element. Two configurations lie in the same S_n -orbit iff they have the same *profile* $(a_t)_t$, i.e., the same number of elements assigned to each position type. The stabilizer of a chosen orbit representative is $\prod_t S_{a_t}$, so $\text{Frob}(\text{Ind}_{\text{Stab}}^{S_n} \mathbf{1}) = \prod_t h_{a_t} = h_\mu$ where μ collects the positive a_t ’s. Summing over profiles and imposing block-nonemptiness gives

$$\text{Frob}(\mathbb{C}[\widetilde{\mathcal{OP}}_{n,r}^{(k)}]) = [x^n] \prod_{i=1}^k \left(\underbrace{\bar{H}^r - 1}_{\text{block } i: \text{ any nonempty color-profile}} \right) \cdot \underbrace{\bar{H}^{r-1}}_{Z\text{-profile}}$$

which is (4). Applying the involution $\omega: h_i \mapsto e_i$ (which corresponds to tensoring with the sign representation, i.e., the det-twist in (1)) turns $\bar{H}$ into $\bar{E}$ and yields the first equality of (5). The binomial expansion $(\bar{E}^r - 1)^k = \sum_{j=0}^k \binom{k}{j} (-1)^{k-j} \bar{E}^{rj}$ combined with $\bar{E}^{rj} \cdot \bar{E}^{r-1} = \bar{E}^{r(j+1)-1}$ gives the second equality. $\square$

3 Power-sum expansion of F_m

Proposition 3. *For all $n \geq 0$ and $m \in \mathbb{Z}_{\geq 0}$,*

$$F_m = \sum_{\mu \vdash n} \frac{\varepsilon(\mu) m^{\ell(\mu)}}{z_\mu} p_\mu. \quad (6)$$

Consequently $m \mapsto F_m$ is a polynomial in m of degree at most n with values in Λ_n , and its coefficient of m^ℓ is P_ℓ .

Proof. Newton’s identity in the plethystic form gives $\log(\prod_i (1 + x_i)) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} p_k$, so

$$\bar{E}^m = \prod_i (1 + x_i)^m = \exp\left(m \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} p_k\right).$$

Expanding the exponential and grouping by the resulting partition $\mu = (1^{\ell_1} 2^{\ell_2} \dots)$,

$$\bar{E}^m = \sum_{\ell_1, \ell_2, \dots \geq 0} \prod_{j \geq 1} \frac{m^{\ell_j} (-1)^{(j-1)\ell_j}}{j^{\ell_j} \ell_j!} p_j^{\ell_j} = \sum_{\mu} \frac{m^{\ell(\mu)} \varepsilon(\mu)}{z_\mu} p_\mu,$$

using $\sum_j \ell_j(\mu) = \ell(\mu)$, $\sum_j (j-1)\ell_j(\mu) = n - \ell(\mu)$ (so the sign is $\varepsilon(\mu)$), and $\prod_j j^{\ell_j} \ell_j! = z_\mu$. Extracting the degree- n part yields (6). The final claim is immediate: $m^{\ell(\mu)}$ has degree $\ell(\mu) \leq n$ in m , and grouping F_m by the exponent ℓ of m gives $F_m = \sum_{\ell=1}^n m^\ell P_\ell$ (the term $\ell = 0$ vanishes for $n \geq 1$). $\square$

4 The subspace W_n and the sign-obstruction

Proposition 4. *For $n \geq 1$, let*

$$W_n := \text{span}_{\mathbb{Q}}\{P_\ell : 1 \leq \ell \leq n\} = \text{span}_{\mathbb{Q}}\{F_m : m \geq 1\} \subset \Lambda_n.$$

Then $\dim W_n = n$. Moreover, for every $r \geq 1$,

$$\text{span}_{\mathbb{Q}}\{\text{Frob}(M_{n,r}^{(k)}) : 0 \leq k \leq n\} = W_n.$$

In particular $W_n^{\text{BR}} = W_n$.

Proof. The partitions of n split by length as a disjoint union $\{\mu \vdash n\} = \bigsqcup_{\ell=1}^n \{\mu : \ell(\mu) = \ell\}$; every length $\ell \in \{1, \dots, n\}$ is realised (by e.g. $(n - \ell + 1, 1^{\ell-1})$), so each P_ℓ is nonzero. Since the P_ℓ 's are supported on disjoint subsets of the p -basis they are linearly independent, giving $\dim W_n = n$.

Proposition 3 gives $F_m = \sum_{\ell=1}^n m^\ell P_\ell$, so $F_m \in W_n$ for every m ; conversely the Vandermonde matrix $(m^\ell)_{\ell, m}$ has rank n on any n distinct values of $m \geq 1$, so $\{P_\ell\}$ lies in the $\mathbb{Q}$ -span of $\{F_m\}$. Thus the two spans coincide.

Finally, for fixed $r \geq 1$, the values $r-1, 2r-1, \dots, (n+1)r-1$ are $n+1$ distinct nonnegative integers. Proposition 2 gives $\text{Frob}(M_{n,r}^{(k)}) = \sum_{j=0}^k \binom{k}{j} (-1)^{k-j} F_{r(j+1)-1}$; the change-of-basis matrix from $(F_{r-1}, F_{2r-1}, \dots, F_{(n+1)r-1})$ to $(\text{Frob}(M_{n,r}^{(0)}), \dots, \text{Frob}(M_{n,r}^{(k)}))$ is the lower-triangular matrix $((-1)^{k-j} \binom{k}{j})_{0 \leq j \leq k}$, which is invertible. Letting k run to n shows the two spans coincide. Hence $\text{span}\{\text{Frob}(M_{n,r}^{(k)})\}_{0 \leq k \leq n} = \text{span}\{F_{r-1}, F_{2r-1}, \dots, F_{(n+1)r-1}\} = W_n$ (the last equality because $n+1$ values of m suffice to span the polynomial-in- m image, cf. the Vandermonde argument above). $\square$

Proof of Theorem 1. By Proposition 4 it suffices to prove $p_{(e^{k'})} \notin W_n$ under the hypothesis $k' \geq 2$, $e \geq 2$, and to exhibit the functional L_ν in (3).

Suppose for contradiction $p_{(e^{k'})} = \sum_{\ell=1}^n \alpha_\ell P_\ell$ for some $\alpha_\ell \in \mathbb{Q}$. Since P_ℓ is supported on partitions of length exactly ℓ and $p_{(e^{k'})}$ is supported at the single partition $(e^{k'})$ (of length k'), we must have $\alpha_\ell = 0$ for $\ell \neq k'$; and $\alpha_{k'} \cdot \sum_{\ell(\mu)=k'} \frac{\varepsilon(\mu)}{z_\mu} p_\mu = p_{(e^{k'})}$. Reading off the coefficient of $p_{(e^{k'})}$ forces $\alpha_{k'} \cdot \frac{\varepsilon((e^{k'}))}{z_{(e^{k'})}} = 1$, so $\alpha_{k'} \neq 0$. Reading off the coefficient of p_ν for any $\nu \vdash n$ with $\ell(\nu) = k'$, $\nu \neq (e^{k'})$, forces $\alpha_{k'} \cdot \frac{\varepsilon(\nu)}{z_\nu} = 0$, i.e. $\alpha_{k'} = 0$ (since $\varepsilon(\nu), z_\nu \neq 0$). This is the desired contradiction *as soon as such a ν exists*, which under our hypotheses is witnessed by $\nu = (e+1, \underbrace{e, \dots, e}_{k'-2 \text{ copies}}, e-1)$:

this has length $k' \geq 2$, all parts ≥ 1 (using $e \geq 2$), is weakly decreasing (using $e+1 \geq e \geq e-1$), sums to $ek' = n$, and is distinct from $(e^{k'})$ because its first part is $e+1 \neq e$.

For the explicit functional, define L_ν by (3). It reads only coefficients at partitions of length k' , hence vanishes on P_ℓ for $\ell \neq k'$; on $P_{k'}$ it evaluates to

$$L_\nu(P_{k'}) = \frac{z_{(e^{k'})}}{\varepsilon((e^{k'}))} \cdot \frac{\varepsilon((e^{k'}))}{z_{(e^{k'})}} - \frac{z_\nu}{\varepsilon(\nu)} \cdot \frac{\varepsilon(\nu)}{z_\nu} = 1 - 1 = 0.$$

Hence L_ν annihilates all of $W_n = W_n^{\text{BR}}$. It evaluates on the target as $L_\nu(p_{(e^{k'})}) = z_{(e^{k'})}/\varepsilon((e^{k'})) \neq 0$. This completes the proof of Theorem 1. $\square$

5 The strong form: sign-ratio detectors

The proof above uses only one functional per required contradiction, but the full annihilator $W_n^\perp$ has dimension $p(n) - n$ and admits a basis of “sign-ratio detectors.”

Corollary 5. *Fix $n \geq 1$. For each length ℓ that supports two or more partitions of n , choose a distinguished partition ν_ℓ^{piv} of n of length ℓ (the “pivot”), and for each other partition μ of n of length ℓ define*

$$L_{\mu, \ell}(\phi) := \frac{z_\mu}{\varepsilon(\mu)} [p_\mu] \phi - \frac{z_{\nu_\ell^{\text{piv}}}}{\varepsilon(\nu_\ell^{\text{piv}})} [p_{\nu_\ell^{\text{piv}}}] \phi.$$

These $\{L_{\mu, \ell}\}$ form a basis of $W_n^\perp$; there are exactly $p(n) - n$ of them.

Proof. Each $L_{\mu,\ell}$ reads coefficients only at length- ℓ partitions, so vanishes on $P_{\ell'}$ for $\ell' \neq \ell$; on P_ℓ it evaluates as $(z_\mu/\varepsilon(\mu)) \cdot (\varepsilon(\mu)/z_\mu) - (z_{\nu_\ell^{\text{piv}}}/\varepsilon(\nu_\ell^{\text{piv}})) \cdot (\varepsilon(\nu_\ell^{\text{piv}})/z_{\nu_\ell^{\text{piv}}}) = 0$. Hence $L_{\mu,\ell} \in W_n^\perp$. Linear independence: the $L_{\mu,\ell}$'s use distinct pairs of p_μ -coefficient-reads, and one verifies by inspection that no nontrivial combination sums to zero. The count is $\sum_\ell (\pi_\ell(n) - 1) = p(n) - n$, where $\pi_\ell(n)$ is the number of partitions of n of length ℓ , which matches $\dim \Lambda_n - \dim W_n$. $\square$

Remark 6. The empirical Schur-basis functionals $L_1, L_2, L_3 \in \Lambda_6^*$ found in the $(n, r) = (6, 3)$ probe (see the accompanying probe log at `probes/2026-08-09-bhattacharya-rhoades-wreath-superspace/RESULT.m`) are $\mathbb{Q}$ -linear combinations of the length-2 sign-ratio detectors:

$$\begin{aligned} L_1 &= \frac{8}{3}[s_{(4,2)}] - 8[s_{(3,3)}] + [s_{(3,2,1)}], \\ L_2 &= -[s_{(5,1)}] + 5[s_{(4,2)}] - 10[s_{(3,3)}] + [s_{(3,1^3)}], \\ L_3 &= [s_{(5,1)}] - \frac{20}{3}[s_{(4,2)}] + 15[s_{(3,3)}] - 5[s_{(2^3)}] + [s_{(2,1^4)}]. \end{aligned}$$

These pair to $-18, -18, +18$ against p_3^2 respectively, in agreement with the sign of the coefficient of the pivot $p_{(3,3)}$ in the target. Computationally we verify that each L_i annihilates all six P_ℓ 's for $n = 6$; see the verification script `verify_probe_L.py`.

6 Verification

The proof rests on three verifiable claims:

- (V1) Proposition 2: $\text{Frob}(M_{n,r}^{(k)}) = \sum_{j=0}^k \binom{k}{j} (-1)^{k-j} F_{r(j+1)-1}$.
- (V2) Proposition 4: $\text{rank span}\{\text{Frob}(M_{n,r}^{(k)})\}_{k=0}^n = n$ for each $r \geq 1$; equal to $\text{rank span}\{F_m\} = \text{rank span}\{P_\ell\} = n$.
- (V3) Theorem 1: $p_e^{k'} \notin W_n$ for $k' \geq 2$, witnessed by an explicit functional.

The scripts in `proofs/2026-08-09-br-work/` confirm all three:

- `verify_gf.py`: for $(n, r) = (6, 3)$ and each $k = 0, \dots, 6$, computes $\text{Frob}(M_{n,r}^{(k)})$ two ways (via BR-orbit enumeration and via the formula) and checks equality in the p -basis. All 7/7 match.
- `verify_span.py`: for each $n \in \{4, 5, 6, 8, 9, 12\}$, computes $\text{rank span}\{F_m\} = \text{rank span}\{P_\ell\} = n$ (all cases pass, with union-span also equal to n); and for each of the 9 test cases $(n, e) \in \{(6, 2), (6, 3), (8, 2), (8, 4), (9, 3), (12, 2), (12, 3), (12, 4), (12, 6)\}$ verifies $p_e^{n/e} \notin W_n$ and produces the explicit L_ν -functional.
- `verify_probe_L.py`: for $n = 6$, confirms each of the empirical L_1, L_2, L_3 from the probe evaluates to 0 on all six P_ℓ 's, and to ± 18 on p_3^2 .

Aesthetic connection to the seed

Bhattacharya–Rhoades's wreath superspace, restricted to S_n , is a generating series of *permutation representations* governed by a compact plethystic-exponential structure ($\bar{E}^r - 1$ per block). The resulting slice-Frobenius family lies in an n -dimensional subspace of Λ_n organised by *cycle length alone*. This is one dimension per possible length, and it is orthogonal to what $q_e^{(n)} = p_{(e^{k'})}$ demands: a delta at a single length- k' partition, in a length class that supports many partitions.

The obstruction is neither a Kostka-cone (nonnegativity) issue nor a multiplicity-2 issue — it is the incompatibility between a *length-graded* single-parameter family (BR side) and a *cycle-type-graded* point (composite- d side). This matches the Levi-stability coordinating-theme observed across 2025–2026: the BR module category is “one length” below the resolution needed. The right level of abstraction for the obstruction is the (only) n -dimensional family $\{P_\ell\}_\ell$, and the sign-ratio detectors capture the codimension- $(p(n) - n)$ complement exactly.

This completes the proof of the Bhattacharya–Rhoades sign-obstruction theorem.

References

- [1] A. Bhattacharya, B. Rhoades. *The wreath superspace and the fermionic-degree filtration*. arXiv:2606.30977, 2026.
- [2] Clio. *Virtual-character sign-pattern theorem for the composite- d symmetric function $q_e^{(n)} = p_e^{k'}$* . Standalone proof, 2026-08-08.